EECE 5639: Computer Vision
Spring 2026, Project 1

Assigned: Thursday, Feb 05, 2026 **Max Points: 100** **Due: Friday, Feb 13, 2026**

Motion Detection Using Image Filtering

Suppose that we have a sequence of images captured with a stationary camera. Most of the pixels belong to a stationary background, and relatively small moving objects pass in front of the camera. The intensity values observed at a pixel over time is constant or slowly varying, *except* when a moving object begins to pass through that pixel. The moving object causes a large change in the pixel intensity over time. Thus, we can detect a moving object by looking into **large** gradients in the *temporal* evolution of the pixel values.

[Code: 50 points] Your team will implement this idea by writing a main program and necessary subroutines as you see appropriate, in MATLAB or Python. Sample videos to test your program are available in the zip files under "Project 1" on Canvas.

NOTE: The grader will run your main program to generate all the relevant numbers and figures. Please organize your code into sections and add sufficient comments such that the grader can follow your organization and logic. Any errors while running your code will result in a significant reduction of points.

The detailed breakdown of tasks and points for your code is described below.

- [**5 points**] The basic outline of your main program should be as follows:
  1. Read in a sequence of image frames and make them grayscale.
  2. As enough frames are available, apply a 1-D differential operator at each pixel to compute a *temporal* derivative.
  3. Threshold the absolute values of the derivatives to create a 0 and 1 mask of the moving objects.
  4. Combine the mask with the original frame to display the results.
- Explore the following variations on the above basic outline.
  1. [**15 points**] For the temporal derivative filter try a simple $0.5[-1, 0, 1]$ filter and a 1D derivative of a Gaussian with a user-defined standard deviation, denoted t_σ. Try at least 3 different values of t_σ in the increasing order. Compare the results of the simple derivative filter and the different cases of the derivative of Gaussian. Discuss your results.
  2. [**20 points**] Try applying a 2D spatial smoothing filter to the frames before applying the temporal derivative filter. For the spatial smoothing filter try 3×3, 5×5 box filters, and 2D Gaussian filters with a user-defined standard deviation, denoted s_σ. Note that this parameter s_σ is different from the previous t_σ. Run the program using at least 3 different values of s_σ, in the increasing order. Discuss your results. Which combination of smoothing and derivative filters yield the best results?
  3. [**10 points**] Vary the threshold to get the mask and see what works best. Design a strategy to select a good threshold for each image. Hint: Most pixels belonging to the background change little and thus their temporal gradients are close to zero. You can model these values as Gaussian zero mean noise and estimate the standard deviation of this noise.

[Report: 50 points] Document your study in a report. Use the IEEE two-column format (https://www.ieee.org/conferences/publishing/templates) or equivalent. The report should be no longer than 8 pages, in 11-point Times New Roman or equivalent. It should include the following:

- **[5 points]** Abstract (a high-level overview of the study and main results
- **[10 points]** Description of Algorithms
- **[10 points]** Experiments and Values of Parameters Used
- **[10 points]** Observation (Include representative images which your team used to test the algorithms, as well as representative outputs showing both "good" and "bad" results.)
- **[10 points]** Conclusions
- **[5 points]** References

Submit one zip file for your team on Canvas. The zip file should contain the entire MATLAB/Python code and a pdf of the report. You do not need to zip any data file (video frame), but please ensure that your code calls the data files by the same names as in the zip files shared under "Project 1" on Canvas.